

## Practical Week 10

### Aim

To investigate Linux inode structures using `ls -li`, `stat`, and `find`, and understand the difference between hard links and symbolic links.

```
navaneeth@Navaneeth: ~/pr: x + v
navaneeth@Navaneeth:~$ mkdir -p ~/practical_week10
navaneeth@Navaneeth:~$ cd ~/practical_week10
navaneeth@Navaneeth:~/practical_week10$ echo "Linux inode practical" > original.txt
navaneeth@Navaneeth:~/practical_week10$ cat original.txt
Linux inode practical
navaneeth@Navaneeth:~/practical_week10$ ls -li original.txt
13688 original.txt
navaneeth@Navaneeth:~/practical_week10$ stat original.txt
  File: original.txt
  size: 22          Blocks: 8          IO Block: 4096   regular file
Device: 8,48      Inode: 13688         Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/navaneeth)   Gid: ( 1000/navaneeth)
Access: 2026-09-27 10:45:49.332245680 +0000
Modify: 2026-09-27 10:45:43.920245126 +0000
Change: 2026-09-27 10:45:43.920245126 +0000
 Birth: 2026-09-27 10:45:43.920245126 +0000
navaneeth@Navaneeth:~/practical_week10$ ln original.txt hardlink.txt
navaneeth@Navaneeth:~/practical_week10$
navaneeth@Navaneeth:~/practical_week10$ ls -li original.txt hardlink.txt
13688 hardlink.txt 13688 original.txt
navaneeth@Navaneeth:~/practical_week10$ stat original.txt
  File: original.txt
  size: 22          Blocks: 8          IO Block: 4096   regular file
Device: 8,48      Inode: 13688         Links: 2
Access: (0644/-rw-r--r--)  Uid: ( 1000/navaneeth)   Gid: ( 1000/navaneeth)
Access: 2026-09-27 10:45:49.332245680 +0000
Modify: 2026-09-27 10:45:43.920245126 +0000
Change: 2026-09-27 10:46:13.084245567 +0000
 Birth: 2026-09-27 10:45:43.920245126 +0000
navaneeth@Navaneeth:~/practical_week10$ echo "Data added through hard link" >> hard
link.txt
navaneeth@Navaneeth:~/practical_week10$ cat original.txt
Linux inode practical
Data added through hard link
```

```
navaneeth@Navaneeth: ~/pr
+
-
□
X

Modify: 2026-09-27 10:45:43.920245126 +0000
Change: 2026-09-27 10:46:13.084245567 +0000
Birth: 2026-09-27 10:45:43.920245126 +0000
navaneeth@Navaneeth:~/practical_week10$ echo "Data added through hard link" >> hard
link.txt
navaneeth@Navaneeth:~/practical_week10$ cat original.txt
Linux inode practical
Data added through hard link
navaneeth@Navaneeth:~/practical_week10$ ln -s original.txt symlink.txt
navaneeth@Navaneeth:~/practical_week10$ ls -li original.txt hardlink.txt symlink.tx
t
13688 -rw-r--r-- 2 navaneeth navaneeth 51 Sep 27 10:46 hardlink.txt
13688 -rw-r--r-- 2 navaneeth navaneeth 51 Sep 27 10:46 original.txt
13690 lrwxrwxrwx 1 navaneeth navaneeth 12 Sep 27 10:46 symlink.txt -> original.txt
navaneeth@Navaneeth:~/practical_week10$ ls -l symlink.txt
lrwxrwxrwx 1 navaneeth navaneeth 12 Sep 27 10:46 symlink.txt -> original.txt
navaneeth@Navaneeth:~/practical_week10$ stat symlink.txt
  File: 'symlink.txt' -> 'original.txt'
  size: 12          Blocks: 0          IO Block: 4096   symbolic link
Device: 8,48      Inode: 13690        Links: 1
Access: (0777/lrwxrwxrwx)  Uid: ( 1000/navaneeth)   Gid: ( 1000/navaneeth)
Access: 2026-09-27 10:47:03.400228905 +0000
Modify: 2026-09-27 10:46:59.580229205 +0000
Change: 2026-09-27 10:46:59.580229205 +0000
Birth: 2026-09-27 10:46:59.580229205 +0000
navaneeth@Navaneeth:~/practical_week10$ stat -L symlink.txt
  File: symlink.txt
  size: 51          Blocks: 8          IO Block: 4096   regular file
Device: 8,48      Inode: 13688        Links: 2
Access: (0644/-rw-r--r--)  Uid: ( 1000/navaneeth)   Gid: ( 1000/navaneeth)
Access: 2026-09-27 10:46:53.648229634 +0000
Modify: 2026-09-27 10:46:49.872231374 +0000
Change: 2026-09-27 10:46:49.872231374 +0000
Birth: 2026-09-27 10:45:43.920245126 +0000
navaneeth@Navaneeth:~/practical_week10$ ls -li original.txt
13688 original.txt
navaneeth@Navaneeth:~/practical_week10$ find . -inum 13688
./hardlink.txt
./original.txt
navaneeth@Navaneeth:~/practical_week10$ |
```

## Observation

- original.txt and hardlink.txt have the **same inode number**.
- The hard link increases the link count.
- symlink.txt has a different inode and points to original.txt.
- find -inum can locate directory entries associated with an inode.

## Result

Linux inode information was successfully examined, and hard links and symbolic links were created and verified using Linux commands.

## Aim

To use the Linux `mmap()` system call for reading and writing a file and understand its difference from traditional `read()` and `write()` operations.

```
navaneeth@Navaneeth: ~/pri x + v
GNU nano 8.7.1 mmap_demo.c *
    close(fd);
    return 1;
}

printf("Original file content:\n");
write(STDOUT_FILENO, mapped, fileInfo.st_size);

memcpy(mapped, "Memory-mapped I/O", 17);

if (msync(mapped, fileInfo.st_size, MS_SYNC) < 0)
{
    perror("msync failed");
}

printf("\nFile modified using mmap().\n");

munmap(mapped, fileInfo.st_size);
close(fd);

return 0;
}|
```

```
navaneeth@Navaneeth: ~/pri x + v
GNU nano 8.7.1 traditional_io.c *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int fd;
    char buffer[1024];
    ssize_t bytesRead;

    fd = open("mmap_input.txt", O_RDONLY);

    if (fd < 0)
    {
        perror("Error opening file");
        return 1;
    }

    bytesRead = read(fd, buffer, sizeof(buffer) - 1);

    if (bytesRead < 0)
    {
        perror("Error reading file");
        close(fd);
        return 1;
    }

    buffer[bytesRead] = '\0';

    printf("File content using read():\n");
    printf("%s\n", buffer);

    close(fd);

    return 0;
}|

^C Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^N Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

## Observation

The file was successfully mapped into memory using `mmap()`. The program accessed and modified the mapped memory, and the changes were synchronized

back to the file using `msync()`. A traditional `read()` program was also implemented for comparison.

## **Result**

**File reading and writing using `mmap()` was successfully implemented and compared with traditional `read()/write()` file I/O.**